MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU – UGP (HONOURS) REGULAR EXAMINATION, APRIL 2026
Second Semester

MG2DSCSTA100– STATISTICS

INTRODUCTION TO STATISTICAL MODELLING

(2024 ADMISSION ONWARDS)

Duration: 1.5 Hours Maximum Marks: 50

Remember (K), Understand (U), Apply (A), Analyse (An), Evaluate (E), Create (C), Interest (I), Appreciation (Ap), and Skill (S)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

[Learning Domain] [CO No(s)]

SCHEME OF VALUATION

Part A

Short Answer Questions

Answer any **seven** questions. Each question carries **2 marks**.

1. Getting pdf as g(y) = $\frac{2(y-1)}{9}$; $1 < y < 4$ – 2 marks [U][1]
2. Definition – 2 marks. [U][2]
3. Two merits or two properties or two applications – 2 marks. Writing only the pdf of Normal distribution – 1 mark [U][3]
4. Mentioning two properties – 2 marks. Definition of pdf only - 1 mark [U][1]
5. Mentioning any two properties – 2 marks . Definition of mgf only - 1 mark [U][2]
6. Answer = 0.1465– 2 marks. Writing pmf of Poisson distribution only-1 mark or writing the parameter value as 4 - 1 mark[U][3]
7. Definition – 1 mark, one example – 1 mark [U][1]
8. Answer = 0.1587 – 2 marks [A][3]
9. Any two properties – 2 marks. Definition of geometric distribution only -1 mark [A][3]
10. Definition- 2 marks [K][3] (7x2=14 marks)

Part B

Short Essay Questions

Answer any **four** questions. Each question carries **6 marks**.

11. Definition - 3 marks ; Any three properties – 3 marks [K][1]
12. Solving and getting answer = 5/7 = 0.714 - 6 marks. Writing formula of E(X) alone – 2 marks [A][2]
13. Getting n = 16 and p = 0.25 - 3 marks and writing the pmf – 3 marks [A][3]
14. Mean = 0.6 and variance = 0.24 - 3 + 3 marks [U][3]
15. Any three properties – 6 marks. Definition only – 3 marks [A][3]
16. Any three properties– 6 marks .Definition only – 3 marks [An][3]

(4x6=24 marks)

Part C

Essay Type Questions

Answer any **one** question. Each question carries **12 marks**.

17. k = 1/30 - 4 marks; $P(X > 1.7) = 29/30$ - 2 marks; $P(1 < X < 3) = 4/30$ - 2 marks; P(X takes an even number) = 4/30 - 2 marks; $P(X \leq 3.4) = 5/30$ - 2 marks [U][1]
18. Writing the pmf of binomial distribution – 3 marks; getting n = 3 and p = 0.8 – 3 marks; (a) answer = 0.384 (b) answer = 0.896 (c) answer = 0.488 (2 + 2 + 2 = 6 marks) or if a student does the problem using direct method and obtain the above same answer, give 4 marks to each sub question (a), (b) and (c). [A][3]